

GrainBrain: Multiview Identification and Stratification of Defective Grain Kernels

Lei Fan , Dongdong Fan, Yiwen Ding, Yong Wu, Donglin Di, Maurice Pagnucco , and Yang Song

Abstract—Grain appearance inspection is crucial for evaluating grain quality and determining seed stratification. Typically, trained inspectors manually examine each grain kernel to identify and remove defective ones, which is time-consuming and error-prone. In this article, we present GrainBrain, a robotic vision-based system comprising a hardware prototype (A100) and a deep learning model (GrainAD). A100 is equipped with five cameras to capture high-quality, multiview images of each kernel. The identification of defective kernels is treated as an unsupervised anomaly detection task. GrainAD trains a classifier to distinguish between healthy and pseudoanomaly samples generated at both image and feature levels, and a supervised contrastive learning loss is employed to obtain compact feature representations of healthy kernels. In addition, we release a large-scale dataset containing over 100K annotated images of four types of cereal grains. Extensive experiments were conducted to verify the superiority of our system, achieving an average AUROC of 94.4/90.4% at the image/pixel level. Our system excelled in both efficiency and consistency, as demonstrated by experiments comparing human experts to the system.

Index Terms—Anomaly detection, deep learning, grain appearance inspection, grain quality.

I. INTRODUCTION

CEREAL grains provide various essential nutrients and serve as fundamental ingredients in food and daily necessities. Grain appearance inspection (GAI) aims at identifying defective grain kernels and is crucial for quality determination in high-throughput grain inspection and seed stratification [1], [2]. GAI plays an indispensable role in various stages of agricultural activities. During breeding, each seed's appearance needs to be carefully inspected to eliminate defective ones that are difficult

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The dataset and code can be found on <https://grainnet.github.io/GrainBrain.html>.

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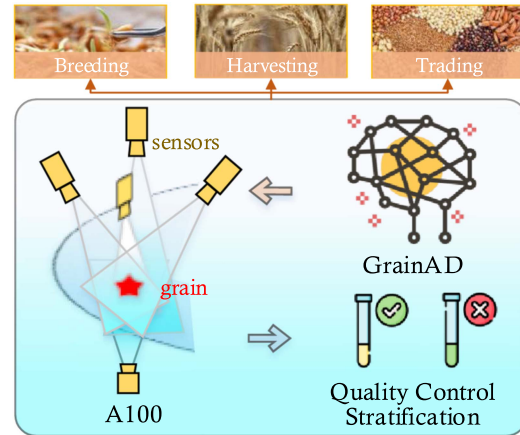


Fig. 1. GAI is crucial for agricultural activities, e.g., breeding, harvesting, and trading. To perform grain quality inspection and stratification, we build a prototype A100 equipped with five cameras to capture multiview images of each grain kernel. An anomaly detection framework GrainAD is developed to identify defective kernels.

to germinate, thereby improving the crop survival rate and yield [3]. At harvest, it reflects the status of pest infestations and Fusarium head blight, acting as a key predictor of annual yield and quality [4]. In grain trading, storage, and processing, it directly estimates the proportion of low-value grain, serving as an important reference for transaction prices [5], as depicted in Fig. 1.

Typically, GAI is conducted manually by professional inspectors following standard ISO procedures [6], [7]. They are required to examine each grain kernel's appearance based on visual characteristics using hand tools (e.g., sieves, tweezers). This process is extremely time-consuming because grain kernels are tiny in size, and the large quantity requires prolonged inspection. Even experienced inspectors need over 25 min to complete the inspection for a set of 50-gram samples. The manual process is prone to human error and inconsistency, which results in inaccurate assessments and variability in inspection outcomes, further complicating the task and reducing the overall reliability. Therefore, it is necessary to develop an automated system for GAI that can capture high-quality visual information of each grain kernel and efficiently filter out defective kernels. However, automated processing and analysis of these grains face several challenges.

- Cereal grains need to be inspected individually, but their small and diverse sizes cause occlusion issues, complicating visual assessment and separation.



Fig. 2. Atlas of four types of cereal grains, including healthy and defective kernels with pixel-level annotations marked by red contours (■).

- Some crucial characteristics (e.g., pest-induced holes) are extremely small, posing challenges for the acquisition environment including cameras and optimal lighting conditions for accurate detection, as illustrated in Fig. 2.
- Different types of grains (e.g., wheat, maize) exhibit high intra- and interheterogeneity. Even experienced inspectors need specialized training due to inconsistencies in inspection standards and personal experience.

To address these challenges, we develop a robotic vision-based system named **GrainBrain**, incorporating a prototype for image acquisition and kernel sorting and a deep learning model for identifying defective grains. Specifically, we comprehensively analyze the characteristics of various types of grain kernels and accordingly build a customized prototype named **A100**. It consists of a feeding cabin, a transparent plate, a capturing module, and a sorting module. The feeding cabin efficiently separates different types of grain kernels one by one. The capturing module is equipped with five high-resolution cameras to capture multiview images of each kernel.

To perform GAI tasks, the identification of defective kernels is formulated as an anomaly detection task, i.e., only normal (healthy) grains are provided as references for training, and models are required to identify defective kernels under an unsupervised learning manner. With this setting, we develop a simple yet effective deep learning model named *GrainAD*. We build upon the SimpNet [8] by converting the unsupervised anomaly detection into a supervised learning manner by leveraging synthetic data. Our model synthesizes pseudoanomalies by injecting noise into normal samples in the image and feature spaces, and then trains a classifier to identify these normal or synthesized data. We further modify the model from both feature and image perspectives. Considering the multiview characteristics of our grain dataset, a supervised contrastive learning (SCL) loss is employed to regularize the feature representations of normal samples by fully exploiting the information shared across different views. To handle the multiview images captured from various views, a spatial transformation network (STN) is introduced to explicitly align the orientations of different views under a self-supervised learning manner. After training, both branches used for synthesizing anomalies are removed, resulting in a streamlined GrainAD that is deployment-ready for integration into the GrainBrain system for real-time inspection.

In addition, we construct a large-scale dataset with over 100K images covering four types of cereal grains: wheat, maize, soybean, and paddy. This dataset includes 21.5K and extra 5.4K defective kernels with image-level and pixel-level annotations, respectively. Extensive experiments were conducted on this

dataset, demonstrating the feasibility and superiority of our system. We further compared our system with human experts in terms of consistency and efficiency. The main contributions are summarized as follows.

- A robotic vision prototype, A100, is designed for capturing high-quality, multiview images of each individual kernel. An anomaly detection model, GrainAD, is developed for the analysis of various types of cereal grains.
- Considering the multiview characteristics of our data, an STN in a self-supervised learning framework is introduced to explicitly adjust a unified pose alignment. We further introduce an SCL loss to regularize the feature representations shared across different views.
- We construct and release a large-scale dataset containing over 100K images with high-quality annotations.

II. RELATED WORK

Grain Appearance Inspection: Inspecting grain appearance is one of the requisite procedures for high-throughput grain quality determination and stratification. According to the ISO cereal vocabulary [6], GAI aims to identify defective kernels, which refer to grains with decreased value, such as those contaminated by *Fusarium* head blight or fungus, crushed by external forces, attacked by pests, or germinated, as shown in Fig. 2. These grains typically have lower nutrient content or negatively impact human health.

Computer vision techniques have been studied in grain inspections for several decades. Limited by sensor technologies and computational resources, early studies [9], [10], [11] attempted to classify subtypes of wheat or impurities based on color distributions, morphological attributes, or textural patterns. Recently, deep learning techniques have achieved considerable progress in smart agriculture [12], [13]. For example, Fan et al. [14] released a large-scale grain dataset related to wheat, maize, and rice, and formulated grain quality inspection as fine-grained recognition, domain adaptation, and out-of-distribution classification. AV4GAInsp [15] incorporated a device with dual cameras to capture visual information.

Unlike previous robotic vision-based approaches [12], [15] that adopt one or two cameras to capture visual information but leave certain areas unobserved, in our GrainBrain system, we design a prototype equipped with five cameras on an elaborate mechanical structure to capture high-quality, multiview images of each kernel, ensuring comprehensive surface coverage of the grains. Importantly, our device not only identifies defective

TABLE I
DETAILED SPECIFICATIONS OF A100

Module	Component	Spec.
Feeding Cabin	Vibrator No.	SDVC34-MR
	Vibrator Model	Electromagnetic
Transparent Plate	Flow Rate ($\dot{n}_f$)	1~10 Kernels/s
	Motor Model	Direct Drive
Cameras	Diameter Size	415 mm
	Angular Velocity (ω_p)	0.63~4 rad/s
Sorting Module	Part No.	MGS130Y-H2
	Frame Rate	46 FPS
	Resolution	1272×1016 pixels
	Focal Length	25 mm
	dots-per-inch (DPI)	0.025 mm/pixel
Sorting Module	Motor Model	42-Step Stepper
	Actuator	Cross-Structure Paddle

kernels efficiently but also performs kernel stratification tasks by separating defective ones individually.

Visual Anomaly Detection: Anomaly detection aims to train models using only normal samples, requiring the model to identify anomalous samples during the test phase. Existing methods [16], [17] can be mainly categorized into the following directions:

Feature-based methods use pretrained models to extract features and characterize the distributions of normal data. DeepSVDD [18], [19] trained a network by minimizing a hypersphere to enclose features of normal data. PaDiM [20] extracted features from image patches and employs multivariate Gaussian distributions for modeling the normal data. Patchcore [21] constructed a memory bank to store multilayer features, detecting anomalies by identifying significant gaps between the test samples and the stored features.

Reconstruction-based methods assume that reconstruction models fail to recover anomalous samples when trained with only normal data. MemAE [22] integrated a memory bank into an autoencoder to store the encoder’s features and retrieve them for reconstruction. UniAD [23] performed feature-level reconstruction and proposes a model capable of one-for-all class anomaly detection. RD4AD [23] enhanced the teacher–student model by using the teacher’s output features as input to the student model.

Synthetic-based methods generate pseudoanomalies using data augmentation, converting unsupervised anomaly detection into supervised classification tasks. Cutpaste [24] synthesized anomalies by cutting an image patch and pasting it at a random location. DRAEM [25] generated anomalies with pixel-level annotations by injecting Perlin noise into normal data. SimpleNet [8] attempted to generate anomalies by injecting Gaussian noise into normal features.

Our captured data includes multiview images of the same object at higher resolutions, requiring the identification of subtle visual differences between normal and defective samples. Feature-based methods depend on aligned data to effectively capture the distributions of patches across different positions, while reconstruction-based models struggle with highly heterogeneous data due to the diverse characteristics of different samples. In contrast, we adopt synthetic-based methods that are well-suited for adapting to various data types by blending noise

into normal data, making them more robust in handling the variability within our dataset. Moreover, the deployed model must ensure fast detection efficiency to maintain the system’s real-time performance. Synthetic-based methods introduce additional branches and synthetic samples to train a discriminative classifier, which can be removed at the testing stage, thus ensuring high efficiency.

III. GRAINBRAIN

Our system consists of two parts: a prototype (A100) and an anomaly detection model (GrainAD), as illustrated in Fig. 3. The prototype is built to efficiently capture high-quality, multiview images of each grain kernel and to sort out defective ones. The model, which follows synthetic-based methods by generating pseudoanomalies at both the image and feature levels, is designed to detect and localize anomalies in multiview images to provide sorting signals for the prototype.

A. Prototype: A100

To emulate the detection habits utilized by human experts, we build a prototype called A100, which efficiently captures the visual information of grain kernels and sorts them based on control signals from GrainAD. A100 integrates four main components: a feeding cabin, a transparent plate, a capturing module, and a sorting module. The detailed specifications of each component are shown in Table I.

The feeding cabin is designed to preprocess a set of test samples to ensure that all kernels are separate and nonoverlapping without occlusion. It is equipped with a high-frequency inductance coil and a rotating motor. Kernels are driven along a track with a preset size of 5 mm and vibrated to separate and move to the transparent plate one by one. The cabin maintains a flow rate of $\dot{n}_f$ grains per second.

The transparent plate is employed to receive grains from the feeding cabin and convey them through the capturing module and eventually to the sorting module. It is circular in shape, connected to a direct drive motor via a flange in the center. The plate rotates counterclockwise within an angular velocity ω_p , controlled by pulses from a servo driver.

The capturing module is equipped with five industrial RGB cameras to capture the superficial information of grains, producing multiview images of each grain kernel. Four cameras are arranged in a quadrilateral pattern, tilted at 25° above the plate, while an additional camera is placed below the plate. The effective camera field-of-view reaches $21.5 \times 27 \text{ mm}^2$. To mitigate the influence of environmental factors (e.g., dust), two sets of LED ring lights are configured to provide sufficient, shadow-free lighting conditions, and the exposure time is denoted as t_e .

The sorting module sorts the grains into different storage boxes based on control signals provided by GrainAD. It consists of a cross-structure paddle and a 42-step stepper motor, with each control action rotating the paddle by 90° to complete a sorting operation.

The entire A100’s efficiency depends on balancing the flow rate $\dot{n}_f$, the plate’s angular velocity ω_p , and the quality of captured images. While increasing the plate’s speed can expedite the grain inspection process, it concurrently reduces camera

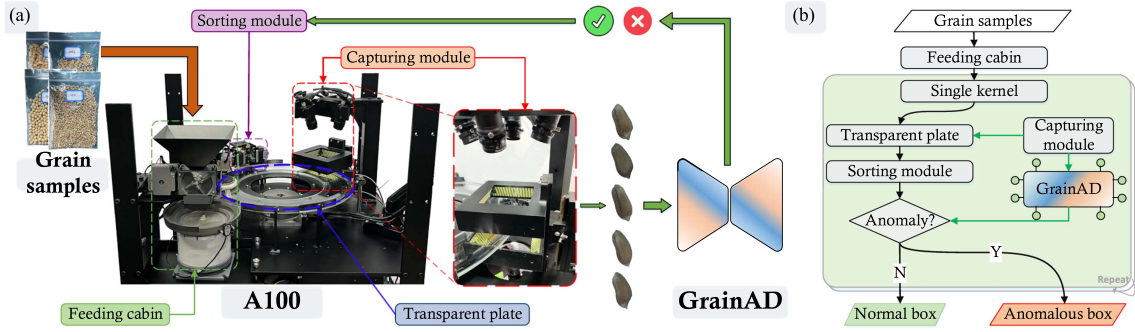


Fig. 3. (a) The diagram of GrainBrain, including a prototype (A100) and a deep learning model (GrainAD). (b) The workflow of GrainBrain, where black lines (■) represent the flow of grain kernels, while green lines (■) indicate the data flow.

exposure times and thereby image quality. This tradeoff is examined in detail in the experimental section. Notably, the processing time of GrainAD must be shorter than the time grains move from the capturing module to the sorting module, ensuring that the grains receive control signals for sorting. Empirically, our prototype can capture 300, 240, 210, and 480 kernel/min for common wheat, maize, soybean and paddy, respectively.

B. Data Acquisition and Dataset Description

Considering that our A100 is designed to accommodate various types of cereal grains, we specifically focus on wheat, maize, soybean, and paddy, as these four types of grains constitute a significant percentage of the world's food supply. To achieve a comprehensive analysis, we collect extensive raw samples from over 30 regions across four countries (CA, CN, KH, and USA). Many of these kernels are preprocessed before being fed into A100 to obtain high-quality images. Specifically, raw grains collected from trucks initially undergo an impurity removal process consisting of two steps. First, two sieves with different mesh sizes are used to filter out excessively large and small impurities. Next, lightweight impurities, such as bran, husks, and straw, may still adhere to the transparent plate due to static electricity, obstructing the cameras from capturing high-quality images. To address this, a low-power fan is employed to remove any remaining impurities.

Data annotations: We engage a team of six experienced inspectors: four senior inspectors with 5–10 years of inspection experience and two experts with over 10 years of experience. We establish a comprehensive procedure. Each kernel involves multiple steps to ensure consistency and accuracy.

- The defective or normal label is confirmed if all four senior inspectors assign the same label.
- If three senior inspectors assign the same label, two experts are consulted to reach a final agreement.
- The remaining images are discarded.

All captured images are re-examined by quality inspectors, and some defective kernels are further provided with pixel-level annotations to indicate anomalous regions.

Dataset: We construct and release a large-scale dataset containing 107.5K images. It encompasses four types of cereal grains with 50, 30, 17.5, and 10K images for wheat, maize, soybean, and paddy, respectively. Each type of cereal grain

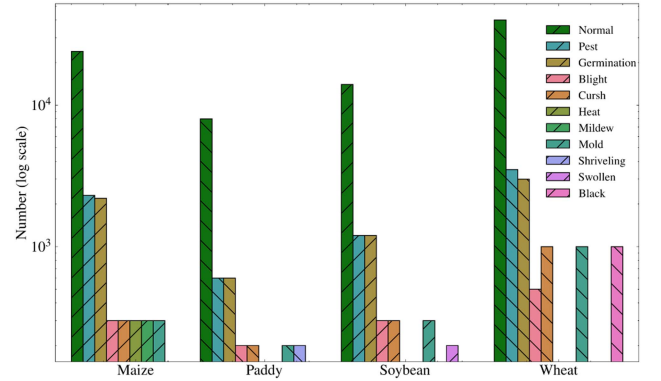


Fig. 4. Data distribution of our released dataset. It contains 107.5K individual kernels covering four types of cereal grains.

includes normal and various numbers of defective kernels. According to ISO standards [6], [7], we made our best effort to collect defective kernels, but it is challenging due to their natural distributions. Detailed information about defective categories and data volume is shown in Fig. 4.

Each image includes five views with a resolution ranging from 510×111 to 3655×797 pixels, where 21.5K defective kernels are categorized into various anomaly categories, and extra 5.4K images of defective kernels are further collected and annotated at the pixel level to highlight defective regions. For example, wheat may exhibit black embryos, while paddy may show shriveled grains without internal kernels. It is noteworthy that there is significant intraheterogeneity among defective kernels within the same category, and many samples simultaneously exhibit multiple defective attributes. This is why we employ anomaly detection methods to train the model instead of the classification task. We believe this categorical information can also be valuable for the community.

C. GrainAD

Preliminary and motivation: We formulate the identification of defective kernels as a classic unsupervised anomaly detection task. Specifically, given a training set $\mathcal{X}_T = \{\mathcal{I}_1^t, \dots, \mathcal{I}_n^t\}$ containing n normal images and a query set $\mathcal{X}_Q = \{\mathcal{I}_1^q, \dots, \mathcal{I}_m^q\}$ including m normal or defective images, each image $\mathcal{I} \in \mathbb{R}^{H \times W \times 3}$ (where H and W are the height and width) has a

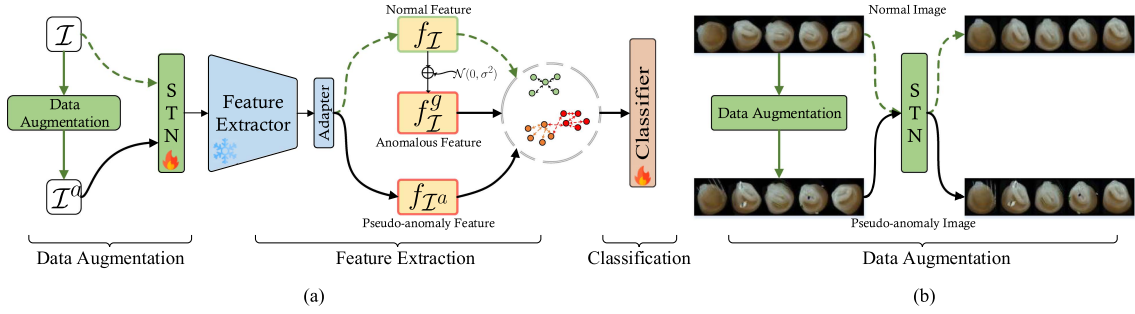


Fig. 5. (a) Architecture of GrainAD. It consists of three stages: data augmentation; feature extraction; and classification. A normal image $\mathcal{I}$ and a pseudo-anomaly image $\mathcal{I}^a$ are fed into the STN, feature extractor, and adapter subsequently to obtain $f_{\mathcal{I}}$ and $f_{\mathcal{I}^a}$, respectively. $f_{\mathcal{I}}^g$ is synthesized by injecting Gaussian noise $\mathcal{N}$ into $f_{\mathcal{I}}$. Then, these features are merged and fed into the classifier. (b) Detailed workflow of data augmentation part. $\mathcal{I}^a$ is synthesized by using data augmentation (e.g., DRAEM [25]) to an input normal image $\mathcal{I}$.

corresponding label $y \in \{0, 1\}$, where 0 and 1 indicate the normal and defective states, respectively. The goal is to train a model using only normal data but to detect defective samples during testing. As the model needs to provide control signals for the sorting module before the corresponding kernel arrives, it must complete the prediction process within approximately 200 ms (based on empirical practical tests). Therefore, the model is expected to be a lightweight and deploy-friendly architecture.

Architecture: We aim to develop **GrainAD** to implement real-time inspection system. As feature-based methods require maintaining a memory bank, reconstruction-based approaches rely on an encoder-decoder architecture. Both methods demand significant computational resources. Therefore, we follow synthetic-based methods that introduce additional branches and synthetic samples during training to train a discriminative classifier, which can be removed at the testing stage, thus ensuring high efficiency in the deployed model. Our framework is primarily based on SimpNet [8] due to its efficiency and simplicity, and it converts the unsupervised problem into a supervised one, in which a classifier is trained to identify anomalies synthesized in both image and feature spaces. It consists of three stages: data augmentation, feature extraction, and classification, as shown in Fig. 5.

In the data augmentation stages, the input image $\mathcal{I}$ is augmented into a pseudoanomaly image $\mathcal{I}^a$ using existing common data augmentation strategies [24], [25]. A spatial transformer network (STN) [26], [27] is employed to regulate the orientation of each view in both $\mathcal{I}$ and $\mathcal{I}^a$. An affine transformation ϕ_θ is employed:

$$\begin{pmatrix} x^t \\ y^t \end{pmatrix} = \phi_\theta = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x^s \\ y^s \\ 1 \end{pmatrix} \quad (1)$$

where ϕ_θ is trained to explicitly learn the spatial transformation from the original coordinates (x^s, y^s) to the target coordinates (x^t, y^t) . It is optimized by minimizing

$$\mathcal{L}_{stn}(I^a, I^b) = \frac{1}{N} \sum_{I^a, I^b \in \mathcal{S}} \|\phi_\theta(I^a) - \phi_\theta(I^b)\|_2^2 \quad (2)$$

where I^a and I^b are two random views of the input image, and a set of $\mathcal{S}$ with N view pairs can be obtained in a batch. Using $\mathcal{L}_{stn}$

encourages ϕ_θ to align the views to a similar orientation. The STN is introduced because trained inspectors typically identify key parts (e.g., the embryo) first and observe that kernels have similar morphological shapes. The primary motivation for using STN lies in its convolutional structure, which is deployment-friendly and can be trained in a self-supervised manner.

In the feature extraction stage, a pretrained model with a feature adapter (denoted as ϕ_e) is employed to extract and transfer multilayer features into the task-specific domain. We use ϕ_e to extract the normal feature $f_{\mathcal{I}}$ and $f_{\mathcal{I}^a}$ from the transformed $\mathcal{I}$ and $\mathcal{I}^a$, respectively. An anomalous feature $f_{\mathcal{I}}^g$ is then generated by adding a Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the normal feature $f_{\mathcal{I}}$. A SCL loss [28] is employed to regularize these features:

$$\mathcal{L}_{scl} = \sum_{i \in \mathcal{V}} \frac{1}{|P(i)|} \sum_{p \in P(i)} -\log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in A(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \quad (3)$$

where $\mathcal{V}$ is the set of all views in the batch. Five views from the same image are treated as individual positive samples $P(i)$ for anchor i , and $A(i)$ is the set of all samples except i . $\mathbf{z}_i$ and $\mathbf{z}_p$ represent the feature representations of samples i and p , respectively, and τ is a temperature parameter. The motivation for introducing $\mathcal{L}_{scl}$ is to ensure that different views of $f_{\mathcal{I}}$ are close to each other while remaining distant from the synthesized features, obtaining more compact feature representations for normal data.

Finally, in the classification stage, the normal and two synthesized features $\langle f_{\mathcal{I}}, f_{\mathcal{I}^a}, f_{\mathcal{I}}^g \rangle$ are concentrated and fed into a classifier τ_c . It is optimized using a binary cross entropy loss:

$$\mathcal{L}_{bce} = -\frac{1}{3} \sum_{i \in \mathcal{F}} y'_i \log(\tau_c(f_i)) \quad (4)$$

where $\mathcal{F}$ denotes the indices of $\{f_{\mathcal{I}}, f_{\mathcal{I}^a}, f_{\mathcal{I}}^g\}$, y'_i is the ground truth label, and $\tau_c(f_i)$ is the predicted probability for class i . The classifier converts the unsupervised anomaly detection into a supervised learning task by generating pseudoanomalies in both image and feature levels. The total training loss is

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{stn} + \lambda_2 \mathcal{L}_{scl} + \lambda_3 \mathcal{L}_{bce} \quad (5)$$

where λ_1 , λ_2 , and λ_3 are weights that balance the contributions of each loss component.

Inference: After training, the branches for generating pseudoanomalies are discarded. The query image $\mathcal{I}^q$ is fed into the STN, feature extractor, and classifier sequentially to obtain the predicted map $s = -\tau_c(f_{\mathcal{I}}^q)$. Anomaly localization is produced by interpolating s , with the anomaly score being the highest value of s .

IV. EXPERIMENTS AND EVALUATIONS

In this section, we introduce the experimental setups, qualitative experiments for A100, experiments for GrainAD, and our GrainBrain system versus human experts experiments.

A. Experimental Setups

GrainBrain: We employed a microcontroller (version STM32F4) running on FreeRTOS operating system to control our prototype, providing unified control signals for optimal performance. An embedded device (Nvidia Jetson Xavier NX) was used as the computational platform to execute our GrainAD model, with all models converted and deployed using the TensorRT framework.

Implementation details: For each subset of our dataset, all defective samples and an equal number of normal samples were included in the test set, with the training and test sets split by 60% and 40%, respectively. For GrainAD, we followed DRAEM [25] and used Perlin noise to generate pseudoanomalies. The feature extractor employed a pretrained WResnet50 [29] and selected layers 2-3 to obtain multilayer features, the feature adapter was a single fully connected layer, and the classifier was a two-layer multilayer perceptron (MLP). For synthesizing feature-level anomalies, σ was set to 0.015 for the Gaussian noise $\mathcal{N}$. For the training strategy, we first trained the STN separately by using only $\mathcal{L}_{stn}$. When using five views together or individually as the input, images were resized to 224×1120 or 224×224 pixels, respectively, and the training batch size was set to 16 with a training epoch of 50. Adam optimizer was employed with a learning rate of $1e-4$. Then, we fixed the STN and the feature extractor to train the adapter and the classifier by setting $\lambda_1 = 0$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.5$. The Adam optimizer was used with a learning rate of $2e-4$, and the batch size was set to 8 with a training epoch of 50. For the evaluation metrics, we followed previous studies [21], [25] and adopted the area under the receiver operating characteristic (AUROC) in both image-level and pixel-level settings, with higher values indicating better performance. The predicted map s of the classifier was smoothed by using a Gaussian filter with $\sigma = 4$.

B. Efficiency of Prototype: A100

The high quality of images captured by the capturing module is crucial for the accuracy of our anomaly detection framework. The image quality is directly related to the specifications of the cameras, the feeding rate $\dot{n}_f$ of the feeding cabin, the angular velocity ω_p of the transparent plate, and the total exposure time t_e of the LED ring light. To ensure the grains from the feeding cabin to the transparent plate remain stable without movement, the feeding rate $\dot{n}_f$ for various types of grains can reach up to

ten kernels per second. Since this rate is significantly lower than the camera's capture frequency (45 FPS), it can be considered insignificant at this stage. We conducted experiments to determine the optimal settings, ensuring high quality while exploring the maximum ω_p to achieve high inspection efficiency. To better evaluate the quality of captured images, we used three dice, white (large), red (medium), and blue (small), with different volumes to simulate various types of cereal grains.

We first fixed the transparent plate in a static condition (i.e., $\omega_p = 0$) and tested four LED exposure times t_e (10, 30, 60, and 100 ms) corresponding to Images-1 through 4, respectively. As shown in Fig. 6, the histograms of Image-1 and Image-4 are focused on low or high intensity, corresponding to underexposure and overexposure conditions. Image-2 and Image-3 have good intensity distributions with most values between 0-25, which aligns well with the fact that the majority of captured images have black backgrounds. From a quantitative evaluation perspective, the mean brightness and contrast of Image-2 and Image-3 are 17.6 and 30.1, showing significant differences from Image-1 and Image-4. High-quality images with proper exposure conditions can be obtained when the exposure time t_e is between 30 and 60 ms.

We further conducted experiments using three dice by fixing the exposure time ($t_e = 45$ ms) and gradually increasing ω_p from 0.8 to 3.2 rad/s. We observed that the visual information could be well captured when ω_p was smaller than 1.6 rad/s, while the medium and small dice experienced motion blurring when ω_p exceeded 2.4 rad/s. This is because the camera's field of view is 21.5×27 mm², and the movement distance s_t (during $t_e = 45$ ms) is approximately 7 to 15 mm according to $s = \omega_p \cdot \frac{1}{2} d \cdot t$ at a higher angular velocity ($\omega_p = 3.2$ rad/s). Empirically, to maintain image quality and inspection efficiency, we controlled the exposure time t_e between 30–60 ms and keep the angular velocity ω_p below 1.6 rad/s.

C. Efficacy of Model: GrainAD

Comparison with SOTA: We compared our GrainAD to classic methods: Patch SVDD [19] and DRAEM [25], advanced methods: Patchcore [21], RD4AD [23], Simplenet [8] and CDO [30], lightweight methods: EfficientAD [31], CRAD [32] and MoEAD [33]. We tested two settings: entire-view ($\cdot^\diamond$) and multiple-view ($\cdot^\dagger$). In the entire-view setting, five views of a single kernel were concatenated and used for training. In the multiple-view setting, each view was treated as a separate input.

As shown in Table II, our GrainAD using the multiview setting consistently outperformed other methods, achieving the highest average AUROC scores of 94.4% and 90.4% at the image-level and pixel-level, respectively. Compared to Patchcore and CDO, which also utilize pretrained models to extract multilayer features, our improvement lies in incorporating the STN to effectively align different views into a unified orientation, facilitating easier feature extraction. In contrast to synthetic-based methods like DRAEM and Simplenet, our use of SCL loss promotes learning compact feature representations of normal data by explicitly considering different views of the same kernels exhibiting similar information. While lightweight models, EfficientAD, CRAD,

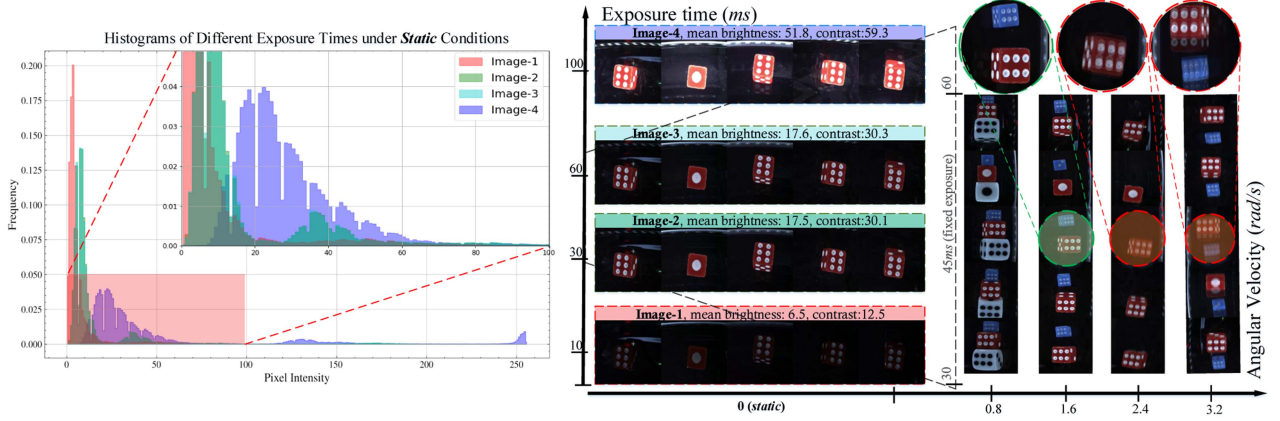


Fig. 6. Examples of captured images with different exposure times t_e and angular velocity ω_p . We first verified the range of t_e with histograms when the plate is static ($\omega_p = 0$). Then, we fixed the exposure time ($t_e = 45$ ms) and further tested different angular velocities.

TABLE II
PERFORMANCE OF GRAINAD IN COMPARISON WITH ADVANCED METHODS

Models	Image-level / Pixel-level AUROC (%)										Computational Complexity		
	Wheat [◇]	Maize [◇]	Soybean [◇]	Paddy [◇]	Average [◇]	Wheat [†]	Maize [†]	Soybean [†]	Paddy [†]	Average [†]	#FLOPS	#Params	#FPS
Patch SVDD [19]	76.1/71.4	67.9/58.5	51.8/71.0	79.2/63.7	68.8/66.2	74.3/86.2	47.0/77.5	83.3/83.1	72.2/83.8	69.2/82.7	58.2G	0.4 M	55.8
DRAEM [25]	68.3/68.7	64.9/50.1	89.7/63.8	81.1/61.6	76.0/61.1	76.8/76.4	78.9/74.5	91.5/79.3	79.1/49.9	81.6/69.9	303.6 G	97.4 M	29
RD4AD [23]	87.3/90.3	79.0/87.7	92.1/89.7	91.6/71.3	87.5/84.8	94.5/93.4	86.1/91.0	92.6/90.3	94.7/76.7	92.0/88.1	51 G	161.1 M	90
Patchcore [21]	87.5/93.2	78.4/86.7	87.1/89.3	94.2/83.9	86.8/88.3	91.3/89.0	87.3/83.9	91.8/87.3	93.9/80.7	91.1/85.2	36.8 G	68.9 M	49
Simplenet [8]	87.1/89.1	79.7/82.1	85.7/83.2	90.6/74.4	85.8/82.2	87.2/84.7	87.0/81.1	93.1/88.9	94.3/78.9	90.4/83.4	51.8 G	72.8 M	84.8
CDO [30]	90.5/93.1	84.6/84.8	88.1/87.9	90.9/70.4	88.5/84.0	90.9/93.2	87.2/86.8	93.3/86.6	92.3/73.7	90.9/85.1	31.7 G	41.4 M	78.2
EfficientAD [31]	71.5/88.4	76.6/79.2	86.4/84.3	90.3/83.6	81.2/83.9	86.8/88.9	76.9/86.1	95.3/88.9	91.1/84.8	87.5/86.9	58.2 G	8.1 M	55.8
CRAD [32]	49.8/72.9	49.4/78.3	49.6/79.5	52.2/75.9	50.2/76.7	62.1/64.8	50.2/71.7	49.3/75.4	53.1/66.1	53.7/69.5	3.7 G	4.5 M	95
MoEAD [33]	81.1/71.5	73.3/68.4	83.8/77.4	79.7/75.4	79.5/73.2	86.5/88.4	77.4/76.8	83.8/79.5	63.4/70.4	77.8/78.8	2.2 G	4.9 M	110.5
Our (GrainAD)	88.3/89.8	85.0/87.6	92.7/89.4	91.7/81.4	89.4/87.1	95.4/94.6	91.6/89.0	96.1/93.5	94.5/84.6	94.4/90.4	52.3 G	124.2 M	80.3

[◇] denotes using the five views of the same image together as the input, while [†] indicates using five views as inputs individually.

and MoEAD employed few conventional blocks to extract features, our model employed WResnet50 as the feature extractor, providing multiscale feature representations with richer spatial and semantic information. However, our model was designed to synthesize pseudoanomalies within relatively small regions, leading to suboptimal performance compared to RD4AD in maize and paddy subsets.

We also observed that many models performed better using the multiple-view compared to their counterparts using the entire-view in both image-level and pixel-level performance. We suggest that the multiple-view approach mimics human inspectors, who carefully examine each view of the grains. Treating each view individually provides higher resolution and more detailed information as input to the model. In the complexity analysis, we observed that RD4AD and Simplenet exhibited similar computational efficiency, but their performance was suboptimal. While CRAD, MoEAD, and EfficientAD had significantly smaller model parameters, their performance varied widely. GrainAD demonstrated the best performance, and its computational efficiency met the requirements for real-time inspection.

Ablation study: We further conducted experiments to verify the effectiveness of each component of GrainAD under the multiview setting, as shown in Table III. Compared to Simplenet, which uses only the synthesized anomaly feature $f_{\mathcal{I}}^g$, integrating the STN to explicitly learn the spatial transformation produced

TABLE III
ABLATION STUDY OF GRAINAD

Settings				Image-level / Pixel-level AUROC (%)				
STN	$\mathcal{I}^a$	$f_{\mathcal{I}}^g$	$\mathcal{L}_{scl}$	Wheat	Maize	Soybean	Paddy	Average
		✓		87.2/84.7	87.0/81.1	93.1/88.9	94.3/78.9	90.4/83.4
✓		✓		93.2/88.1	83.6/82.3	95.6/86.3	95.3/81.5	91.9/84.5
✓	✓			90.3/90.5	89.5/88.0	95.8/93.0	90.0/66.0	91.4/84.4
✓	✓	✓		93.1/88.7	90.9/88.2	95.2/89.5	93.4/86.4	93.1/88.2
✓	✓	✓	✓	95.4/94.6	91.6/89.0	96.1/93.5	94.5/84.6	94.4/90.4

improvements in average image-level and pixel-level AUROC of 1.5% and 1.1%, respectively. We suggest that adding an STN results in poorer performance on maize because it lacks a distinct shape pattern compared to other types of grains, making it difficult to normalize to a similar shape. Using only the pseudoanomaly image ($\mathcal{I}^a$) achieved similar performance to using only the synthesized anomaly feature $f_{\mathcal{I}}^g$, and combining them yielded an average improvement of 2.7% and 4.8% in image-level and pixel-level AUROC, respectively. Employing $\mathcal{L}_{scl}$ allowed the model to achieve the best performance in average scores, obtaining the highest results in wheat, maize, and soybean.

Ablation study for the loss function: We conducted experiments to explore different settings of weights λ_1 , λ_2 , and λ_3 to balance the losses $\mathcal{L}_{stn}$, $\mathcal{L}_{scl}$, and $\mathcal{L}_{bce}$, respectively, as shown in Table IV. Note that the STN in all models was pretrained to convergence. We observed that different weight settings

TABLE IV
ABLATION STUDY OF WEIGHTS FOR THE TRAINING LOSS (REPORTED USING IMAGE-/PIXEL-LEVEL AUROC (%))

λ_1	λ_2	λ_3	Wheat	Maize	Soybean	Paddy	Average
0	0	1	93.1/88.7	90.9/88.2	95.2/89.5	93.4/86.4	93.1/88.2
0	0.5	0.5	95.4/94.6	91.6/89.0	96.1/ 93.5	94.5/84.6	94.4/90.4
0	0.5	1	92.0/94.0	92.3/89.0	96.1/91.7	93.0/86.7	93.4/ 90.4
0	1	0.5	93.1/92.8	91.1/85.0	95.9/91.7	93.6/86.1	93.4/88.9
0.5	0	1	94.2/ 95.3	91.4/86.7	96.1/89.8	93.3/83.8	93.7/88.9
0.5	0.5	1	93.4/95.0	92.1/89.4	95.8/90.3	91.4/ 87.2	93.2/ 90.4
1	0	0.5	94.0/95.1	92.0/87.0	96.0/89.9	93.7/82.1	93.9/88.5
1	0	1	93.5/94.9	91.5/ 90.7	95.7/89.4	92.0/86.2	93.2/90.3
1	1	1	93.0/93.9	89.4/85.0	95.7/93.3	90.7/85.7	92.2/89.5
2	2	1	93.6/94.8	91.7/85.9	95.9/82.5	92.6/84.0	93.5/86.7

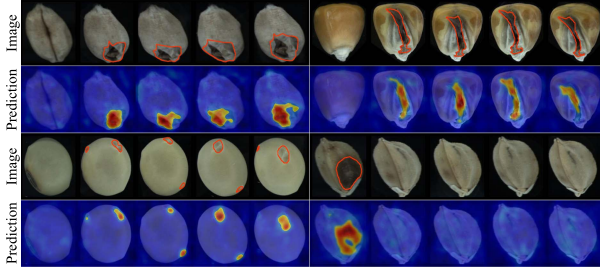


Fig. 7. Visualizations of the pixel-level predictions of our GrainAD. The experts' annotations are highlighted using red contours (■).

produced similar performance. Without explicitly fine-tuning $\mathcal{L}_{stn}$ (i.e., setting λ_1 to 0), the models achieved relatively better performance across four types of cereal grains, and the best performance was obtained when λ_2 and λ_3 were both set to 0.5. We suggest that the STN should be trained in the early stages using self-supervised learning to achieve stable pose alignment. Then, only feature alignment and classification losses should be used for optimization.

Visualization: We also visualized the pixel-level predictions of our GrainAD, as illustrated in Fig. 7. It was observed that our GrainAD effectively localized the anomalous regions of defective kernels. Particularly, when the query kernel is attacked by pests or is in a germination state, our model precisely highlighted the pest-affected areas and the germination sites across multiple views. The pixel-level results significantly supported the model's interpretability and transparency, especially in inspecting various types of grains. We further conducted a quantitative analysis of the effectiveness of STN, as shown in Fig. 8. Without STN, the models demonstrated good pixel-level results for identifying anomalous regions but produced several erroneous predictions. With STN, it successfully aligned the grains' pose to a similar orientation in an unsupervised setting, and the aligned images were used as input for the models, significantly reducing erroneous regions.

D. GrainBrain Versus Human Experts

The GrainBrain is developed to assist inspectors in reducing workloads, especially when dealing with large quantities of grain kernels for quality inspection and stratification tasks. Our goal is to create a system that maintains high consistency with human experts while ensuring high efficiency. To evaluate this, we conducted a comparison between GrainBrain and human experts.

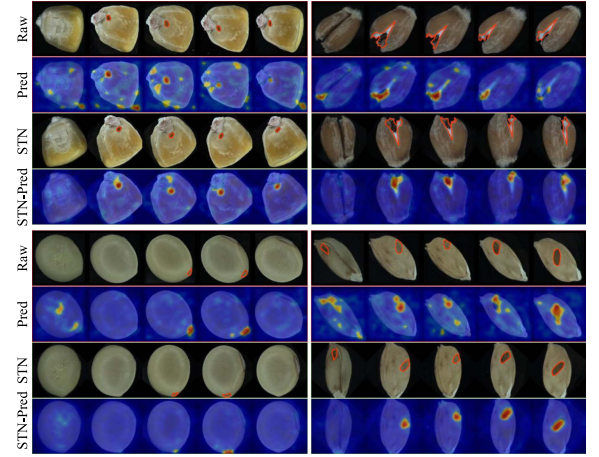


Fig. 8. Visualizations of the pixel-level predictions with/without using STN. The experts' annotations are highlighted using red contours (■).

TABLE V
KAPPA TESTS BETWEEN GRAINBRAIN AND HUMAN EXPERTS

	Cohen's Kappa κ (%)				
	Wheat	Maize	Soybean	Paddy	Average
D_1 vs. H_1	71.5 $\pm$ 4.4	58.1 $\pm$ 3.1	64.8 $\pm$ 2.5	67.1 $\pm$ 4.5	65.4 $\pm$ 3.6
D_1 vs. H_2	72.4 $\pm$ 1.4	60.9 $\pm$ 3.4	64.7 $\pm$ 3.7	64.5 $\pm$ 3.3	65.6 $\pm$ 3.0
D_2 vs. H_1	66.7 $\pm$ 3.1	56.8 $\pm$ 4.6	65.9 $\pm$ 3.2	67.7 $\pm$ 2.4	64.3 $\pm$ 3.3
D_2 vs. H_2	73.8 $\pm$ 3.6	55.6 $\pm$ 2.4	67.3 $\pm$ 3.5	65.8 $\pm$ 2.9	65.6 $\pm$ 3.1
	Cost Time (s/per test sample)				
	Wheat	Maize	Soybean	Paddy	Average
D_1	168 $\pm$ 3	41 $\pm$ 2	137 $\pm$ 8	257 $\pm$ 4	151 $\pm$ 4.5
D_2	179 $\pm$ 9	52 $\pm$ 8	147 $\pm$ 6	286 $\pm$ 7	166 $\pm$ 7.6
H_1	3686 $\pm$ 94	611 $\pm$ 91	1478 $\pm$ 115	5566 $\pm$ 294	2835 $\pm$ 149
H_2	3666 $\pm$ 115	606 $\pm$ 103	1473 $\pm$ 76	5571 $\pm$ 278	2829 $\pm$ 143

Specifically, we built two versions of GrainBrain (denoted as D_1 and D_2) and recruited two human experts (denoted as H_1 and H_2) with over 5 years of specialized experience. We prepared 16 additional 50-gram test samples, with 4 samples for each type of cereal grains. All samples were randomly acquired from grain delivery trucks and underwent an impurity removal process using a series of tools, such as sieves and a fan. The proportion of defective kernels in each test sample ranged from 5–20%. To ensure a fair comparison, experts were required to inspect images captured by our systems rather than real grain kernels. The samples were shuffled and inspected under a dual-blind policy. We conducted the Kappa test to assess agreement and reported the Cohen's kappa value κ and the inspection time using average values and standard deviations. It is noted that when $\kappa \geq 0.6$, it indicates significant agreement, while a κ of 0 indicates agreement by chance.

As shown in Table V, GrainBrain systems demonstrated strong consistency with human experts, achieving Cohen's Kappa values κ of nearly 65% for all comparison experiments across various types of cereal grains. These results validated the feasibility and potential of our system for visual inspection and seed stratification tasks. It is evident that our system demonstrated a significant advantage in inspection time, costing near 174, 46, 142, and 271 s per test sample for wheat, maize, soybean, and paddy, respectively, surpassing H_1 and H_2 by over ten times. Especially when processing wheat or

paddy, which have smaller physical sizes, our systems were faster by approximately $20\times$ speedup. These results verified the efficiency of our system. In practice, inspectors typically take more time because they need to manipulate the grain kernels using tweezers and individually pick, classify, and tally the corresponding proportions.

V. CONCLUSION

In this article, we presented a multiview vision-based system: GrainBrain for high-throughput grain appearance inspection and seed stratification tasks. It consists of a prototype A100 and an anomaly detection model GrainAD. Specifically, A100 is designed to separate kernels individually and capture five-view visual images of each grain kernel efficiently. GrainAD is developed to analyze the captured images and provide control signals for A100 to implement kernel sorting. Our GrainBrain offers automatic detection, inspection, and individual separation of defective kernels. Built upon an anomaly detection framework, our approach significantly enhanced model robustness, enabling the effective detection of various types of anomalies, including unseen or unknown defects. In addition, we released a large-scale dataset containing over 100K images with high-quality annotations.

Our system still has some limitations in both the prototype and algorithms. On the prototype side, the placement of the cameras restricts the ability to capture certain areas of elongated grains, such as maize, or grains with recessed surfaces, resulting in some regions being left unobserved. In addition, different grains, such as wheat and maize, exhibit varying reflectivity under the same lighting conditions. Currently, we used the same exposure time for all grain types, but dynamic adjustments based on the specific grain type should be considered. The fixed focal length of the cameras also limits the ability to capture high-quality images of agricultural products of varying sizes, such as peanuts or millet. On the other hand, the performance of the unsupervised anomaly detection method has not yet reached the level of high expert consensus. Incorporating a small number of labeled anomaly samples as auxiliary data or leveraging large language models could further enhance the model's performance while maintaining robustness. We hope that GrainBrain can stimulate and draw more attention to various fields of grain quality inspection.

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